

TouchLog: Finger Micro Gesture Recognition Using Photo-Reflective Sensors

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ABSTRACT

Fingertip input allows for interactions that are natural, easy to perform, and socially acceptable. It also has advantages in terms of low physical demand, confidentiality, and haptic feedback. In this study, we propose TouchLog, a fingernail-type device that uses skin deformation of the fingertip to identify finger micro gestures written with the thumb on the index finger. TouchLog is attached to the index fingernail and allows for one-handed fingertip input without compromising the haptic feedback on the finger. To evaluate the accuracy of 11 types of finger micro gesture recognition, we conducted a user study ($N = 10$) and obtained an average identification accuracy of 91.5% ($SD = 3.1\%$). A continuous input method using skin deformation and contact pressure was also examined, and its usefulness as a wearable device was discussed.

CCS CONCEPTS

• Human-centered computing → Gestural input.

KEYWORDS

skin deformation, photo-reflective sensor, finger micro gesture, fingernail-type device, wearable system

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1 INTRODUCTION

Since human fingertips are dexterous and capable of precise movements, input using fingertips is expected to be intuitive and diverse. Among fingertip gestures, the finger micro gesture of writing letters on the index finger with the thumb [3, 6, 34] is considered natural, interactive, and socially acceptable [29].

Compared with the mid-air gestural input methods [9, 12], finger micro gestures are advantageous in terms of providing haptic feedback and reduced physical demand. In addition, they introduce more confidential input methods that use the entire hand or wrist [5, 14, 32], as minute movements in the fingertips as input actions.

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Figure 1: TouchLog detects finger micro gestures when the thumb is placed on the index finger.

Furthermore, compared with the voice input method, it can be used with no concern for environmental noise. Thus, there are many advantages to using finger micro gestures.

Various types of finger micro gesture recognition devices have been proposed, including glove-type, ring-type, and fingernail-type devices. Of these, ring-type devices, which are worn on the index finger [1, 3, 6, 10, 16, 28], and fingernail-type devices, which are worn on the fingernail [2, 14, 16], are practical because they provide natural input behavior when worn on the index finger, and these systems are more compact and socially acceptable than the glove-type systems [9, 22, 30].

Ring-type devices are understood as natural accessories for continuous as well as discrete input. However, it is not possible for these devices to acquire contact pressure during their user's gestures. Contact pressure can be measured and used as a new input. In addition, Tiptext and FingerT9 also can realize highly accurate and fast text input by attaching a sensor to the finger pad [32, 34]. However, since the fingertip plays an important role in recognizing the shape and texture of an object, attaching a sensor to the finger pad would allow the tactile sensation of the fingertip to change.

In this study, we proposed a one-handed fingertip gesture input method that does not cover the finger pad. We implemented a fingernail-type device called TouchLog (Figure 1), which is attached to the index finger and detects skin deformation when the index finger and thumb make contact using seven photo-reflective sensors. After gathering sensor data, we extracted features from the acquired time series data, and gesture identification was performed using random forest classifier.

Eleven finger micro gestures were used for evaluation, and 10 fold cross-validation was performed on 10 users, resulting in an average identification accuracy of 91.5% ($SD = 3.1\%$). In addition, the user dependence of the device was examined. Furthermore,

there is a possibility of continuous input through skin deformation and contact pressure, and this was discussed. Since this method uses a fingertip device with a photo-reflective sensor as a gesture recognition system, it is possible to realize a low-cost and compact device. In addition, privacy is ensured since no camera is used. Also, unlike research [6], which uses a variety of sensors and devices to perform recognition, a single device can be used to complete the identification.

The contributions of this study are as follows.

- We propose a fingernail-type device (TouchLog) that allows one-handed input without the loss of tactile sensation.
- Through a user study, we show that our device achieves 91.5% (SD = 3.1%) average identification accuracy when recognizing 11 different gestures.
- The possibility of continuous and discrete input as well as input using contact pressure was discussed.

2 RELATED WORK

2.1 Fingertip input method

Input methods using the fingertip of one hand have been proposed using various sensors and machines, including millimeter wave radars [17], cameras [1, 26], microphone and optic sensors [33], and capacitance sensors [32, 34]. Among them, the approach using a capacitance sensor is easy to use and confidential because the sensor is attached to the finger pad for input. However, the tactile sense of touch in the finger pad, which plays an important role in recognizing the shape and texture of objects, would be impaired.

Many ring-type devices have been proposed as systems that do not compromise the tactile sensation of the finger pad. Various sensors such as front-facing passive infrared (PIR) sensors [6], electric field sensing [3], active electrical sensing [10], and Inertial Measurement Unit (IMU) [28] have been used to identify minute changes in the fingertip. Li et al. [16] also proposed a gesture identification method that made use of a physiological feature of the skin: the change in fingertip color distribution due to fingertip pressure.

In addition, a fingernail-type device that does not cover the finger pad has been proposed. By placing magnetic or capacitive sensors on the fingernail, interactive micro gestures of the fingertip can be recognized [2, 14].

In this study, a photo-reflective sensor is mounted on the fingernail side to enable one-handed input without compromising the tactile sensation of the finger pad. In addition, as with TipText [34], the input action is natural and as simple as touching the index finger with the thumb.

2.2 Sensing technology using photo-reflective sensors

Photo-reflective sensors consist of a light emitting element that emits infrared light and a phototransistor-light-receiving element. The light is emitted from the light emitter and reflected by the object to be measured; then, it is detected by a phototransistor. The distance is measurable because the amount of light reflected varies depending on the distance between the object and the sensor.

As photo-reflective sensors are inexpensive and easily miniaturized, they are used in various sensing applications [13, 20, 21, 27]. In particular, many studies using photo-reflective sensors focused

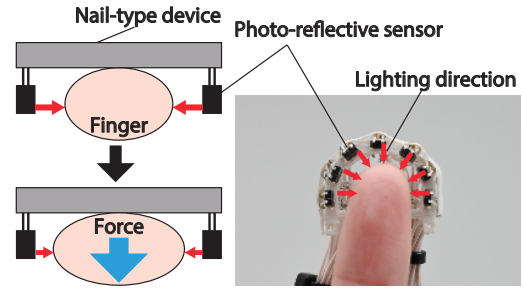


Figure 2: Principles of skin deformation (left) and direction of infrared irradiation (right).

on measuring skin deformation. Mask-type devices [15], eyewear-type devices [7, 18], and ear accessories [4] that measure facial skin deformations to identify facial expressions or face-related gestures have been proposed. The detection of skin touches by hands has also been proposed, such as in earphone-type devices to detect the ear-touching gestures [11], a gestural input method involving face rubbing [19], or touching gestures on cheek skin [35]. Ogata et al. proposed an input method that involves skin deformation when the finger touches the forearm, as well as a method for measuring finger rotation based on skin deformation caused by a ring attached to the index finger [23, 24]. As to the skin deformation of the fingertip, Saito et al. used a photo-reflective sensor to measure the skin deformation of the fingertip and proposed a method for estimating the contact force of the fingertip [25].

In this study, we use a photo-reflective sensor to measure skin deformation when the thumb and index finger touch and to identify finger micro gestures of the fingertips. The present study shares similarities with the aforementioned studies in that the sensor measures skin deformations and identifies gestures.

3 IMPLEMENTATION

When force is applied to the skin of the fingertips, skin deformation occurs (Figure 2 (left)). The degree of deformation varies depending on the force applied, and the amount of skin deformation varies depending on the gesture performed. In this system, a photo-reflective sensor is placed on the nail to acquire skin deformation information. In the following sections, this system is described in terms of both hardware and software.

3.1 Hardware

Fingertip skin deformation information is acquired using a fingernail-type device. This nail is made via 3D printer, and seven photo-reflective sensors (SG-105) are placed around the finger (Figure 2 (right)). The photo-reflective sensors are connected to a micro-controller (Arduino Pro Mini 3.3V) that sends sensor data to a PC via serial communication. The device and the finger were attached using tape.

3.2 Software

Time series data on sensor values for each gesture was acquired from the photo-reflective sensors. The sampling rate in this experiment was about 66 [fps], and 160 frames were acquired; therefore,

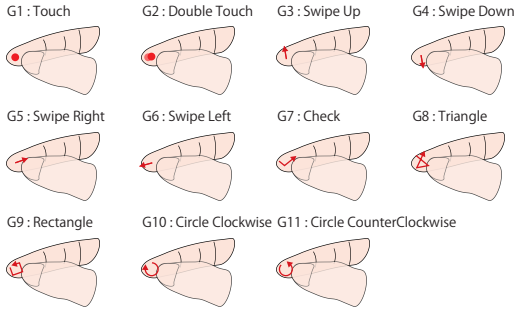


Figure 3: Set of eleven finger micro gestures

each gesture took about 2.4 seconds. A certain threshold was determined, and the gesture was initiated when the threshold was exceeded. The sensor data at the start of the gesture was recorded, and the difference between sensor data at the start and the acquired sensor data was obtained.

Next, the acquired time-series data was divided into five parts (with 32-frame intervals), and statistical features were extracted for each divided part. The mean, median, maximum, minimum, and variance were used as statistical features. Feature extraction was performed using seven photo-reflective sensors, and a feature value of 175 dimensions (for 7 sensor values $\times$ 5 divisions $\times$ 5 statistical features) were extracted for each gesture. The data from which the features were extracted was divided into test and training data, and a random forest was used to identify gestures. A random forest classifier was created using the scikit-learn library in python.

4 EVALUATION

4.1 Gesture sets

Eleven types of finger micro gestures were used in this study (Figure 3). Nine types (Touch, Double Touch, Swipe Up, Swipe Down, Swipe Right, Swipe Left, Check, Circle Clockwise, Circle Counterclockwise) were selected with reference to the fingertip ring-based gesture study [3]. To ensure diversity of gestures, in addition to the manipulative gestures, we also selected two types (Triangle, Rectangle) with reference to finger micro gesture research on fingertips and non-stroke gestures useful that were in touch screen devices [6, 31].

4.2 Procedure

We conducted a user study to compile a dataset of 11 finger micro gestures and evaluate identification accuracy. A total of 10 participants (5 males and 5 females; all 10 were right handed) with a mean age of 22.4 (SD = 1.57) participated in the experiment. All participants wore the device on their right index finger. Before the experiment, each gesture was practiced to ensure that all gestures could be properly performed by the participants. 20 measurements were taken for each gesture, with a break occurring after 10 measurements. If the writing order was incorrect, a re-measurement was performed. The experiment was conducted indoors.

To consider device usage (personal or shared devices) and user dependency, three models (General, Individual, and Leave One Out Cross Validation (LOOCV)) were created and evaluated for their

average identification accuracy. In the General model, we used data from all users and performed a 10-fold cross-validation with a 9:1 split between training and test data. In the Individual model, a 10-fold cross-validation was performed for all users, using data from one user and splitting the training and test data 9:1. In the LOOCV model, data from nine users were used as training data, and the remaining one user was used as test data. The average identification accuracy was calculated for 10 different combinations so that all users were used as test data.

4.3 Result

The mean identification accuracy of the General model was 91.1% (Figure 4 (a)), the mean identification accuracy of the Individual model was 91.5% (SD = 3.1%, Figure 4 (b)), and the mean identification accuracy of the LOOCV model was 67.6% (SD = 14.6%, Figure 4 (c)).

The gestures that were difficult to identify in the General and Individual models were triangles and circle clockwise as well as rectangles and circle counterclockwise, respectively, indicating that it was difficult to recognize the difference between curves and straight lines. These gestures were not recognized properly because the writing direction was the same, although there were differences between straight lines and curves.

The range of accuracy in the LOOCV model was 40.9–88.2%, and the accuracy varied greatly depending on the user who was used as test data. In addition, when compared to the accuracy calculated by the Individual model, the results were low, indicating a high degree of user dependence on the device. Therefore, the current system requires individual calibration before use. In addition, the accuracy of the General model indicates that it could be used as a shared device if calibrated.

5 DISCUSSION, LIMITATIONS & FUTURE WORK

5.1 Other input possibilities for TouchLog

Additional experiments were conducted to investigate the possibility of other TouchLog inputs. The experiment had one participant (1 male; 1 was right handed) with a age of 22, and the experimental environment and feature extraction were the same as described in the Implementation section. Ten frames of time-series data for 4 different gestures (Up, Down, Right, and Left) were acquired. A total of 10 sets of measurements were taken, and a 10-fold cross-validation was performed. The mean identification accuracy was 87.5%. Real-time estimation was performed with the model created at this time, and the pointer was moved up, down, right, and left (Figure 5 (a)(b)).

In addition, we considered the use of contact pressure as an input. By using a photo-reflective sensor and acquiring skin deformation information, contact pressure at the fingertips and foot pressure can be estimated [8, 25]. We believe that by using contact pressure as an input, haptic feedback can be obtained, and intuitive input and the dimensional expansion of the input space can be realized. In this study, fingertip contact pressure was measured and estimated in real time using the same method used in the study of Saito et al. [25]. The regression model was created based on the correlation between fingertip skin deformation and true values of contact pressure [N]. The true value of the fingertip contact pressure was measured with

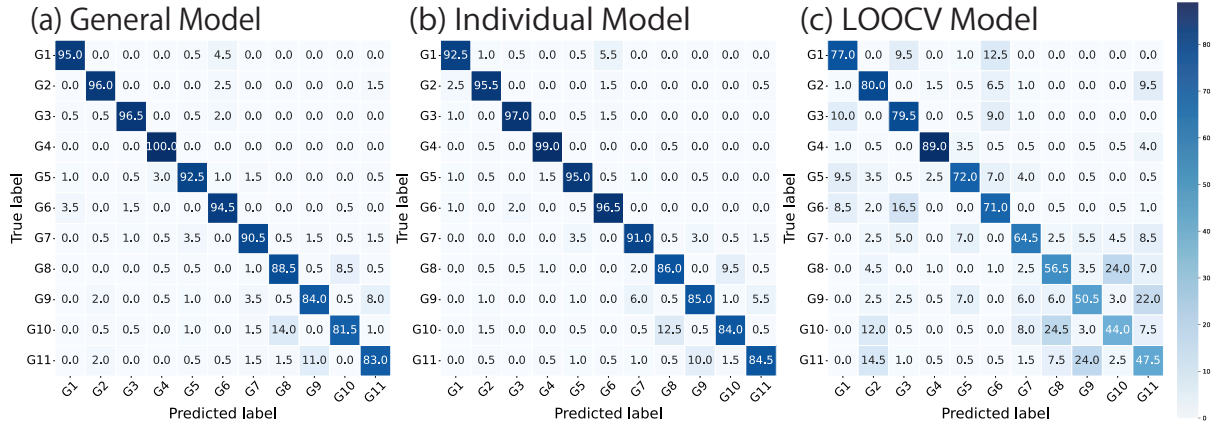


Figure 4: Accuracy of each gesture with different models. (a) General model; (b) Individual model; (c) LOOCV model.

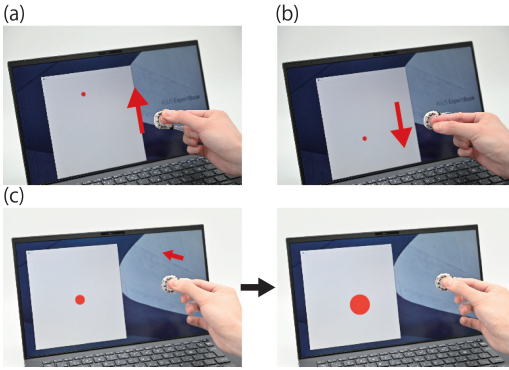


Figure 5: (a) Continuous input (Up); (b) Continuous input (Down); (c) Input by contact pressure

a load cell (TecGihan Co., Ltd.). Ten sets of measurements were taken (1,800 frames were acquired for each set), and the coefficient of determination was 0.51. The estimated results are reflected in the size of the pointer, allowing for input perpendicular to the finger pad (Figure 5 (c)).

Thus, TouchLog has the potential to support discrete as well as continuous gesture input. We will conduct user studies with continuous input to verify whether it can be used effectively.

5.2 Individual differences

TouchLog's high user dependence may be influenced by finger size, dexterity, and fingertip skin condition.

One device was used in this experiment. Since there are individual differences in finger size, the drawing space of the gesture changed depending on the size of the finger. Therefore, several different sizes of devices must be made available.

In addition, since this method uses sensor difference value data, we believe that the influence each individual user is minimal. However, even within the same gesture, the amount of skin deformation caused by the gesture, the speed of the gesture, and the politeness

with which the gesture is written vary from user to user. Furthermore, dexterity may contribute significantly to user dependence, since finger micro gestures have a small drawing space.

It is also possible that the skin condition of the fingertips of the same experiment participants may vary depending on the season and weather. The effects of humidity and fingertip softness were not investigated.

Future work will investigate the effects of device size, dexterity, and fingertip skin condition on input.

5.3 Use in a real environment

Photo-reflective sensors measure the reflected light of irradiated infrared rays and are therefore affected by sunlight. It is known that sensor values can differ when measurements are taken in locations that are easy for sunlight to reach even indoors, such as near a window. Since this experiment was conducted indoors, where sunlight is difficult to reach each time, the effect was considered small. In addition, by measuring the difference from the initial value for each measurement, we made an effort to reduce the influence of ambient light. However, the sensors may be affected by sunlight when used outdoors.

In this experiment, we instructed the participants not to move their hand with the device attached as much as possible during the experiment. Therefore, we did not consider the use of the device while the user is walking or running.

In the future, we plan to conduct verification under conditions similar to real-world environments, such as outdoors or while walking, to ensure practicality. Furthermore, we will examine the contribution of sensors, reduce the number of sensors, adjust the size to fit each user, and make the device wireless, aiming for a device that can be worn comfortably in daily life.

6 CONCLUSION

In this paper, we investigated whether finger micro gestures can be identified by acquiring skin deformation information when the thumb touches the index finger. The average accuracy was 91.5% (SD = 3.1%) for 10 participants. We also investigated the possibility

of continuous input using a fingernail-type device. We believe that this method is useful as a wearable device because it allows discrete and continuous input using skin deformation information.

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